Class 05: Data Visualization with GGPLOT

Jiselle Arevalo

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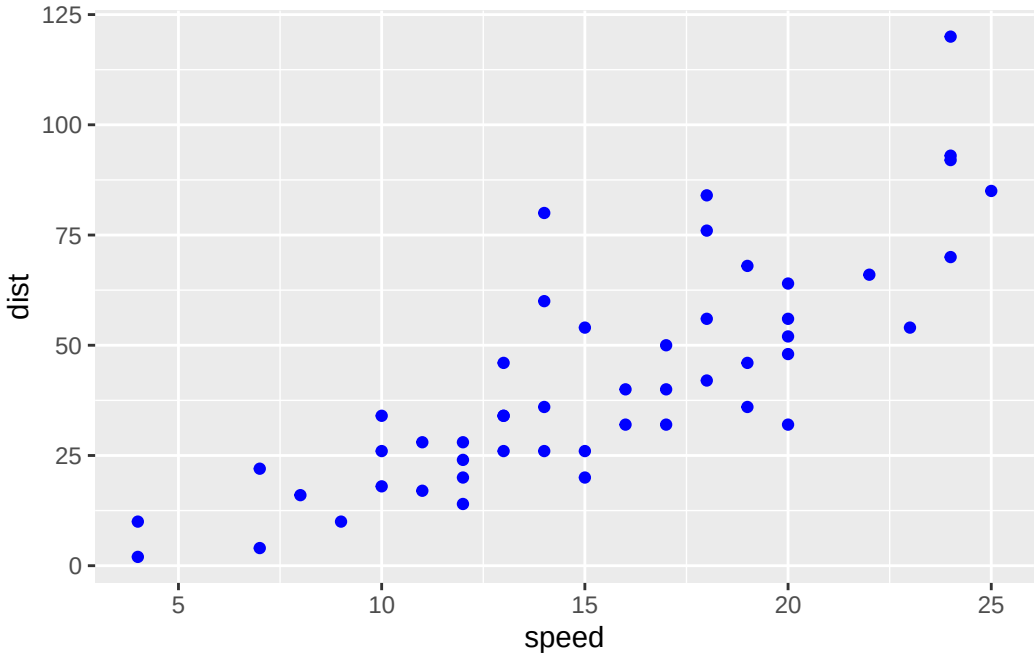
> text
>
> **Q1: Make a ggplot plot**
>
> **Answer Q1: hshshd**

```
x <-1:5
x + 1
```

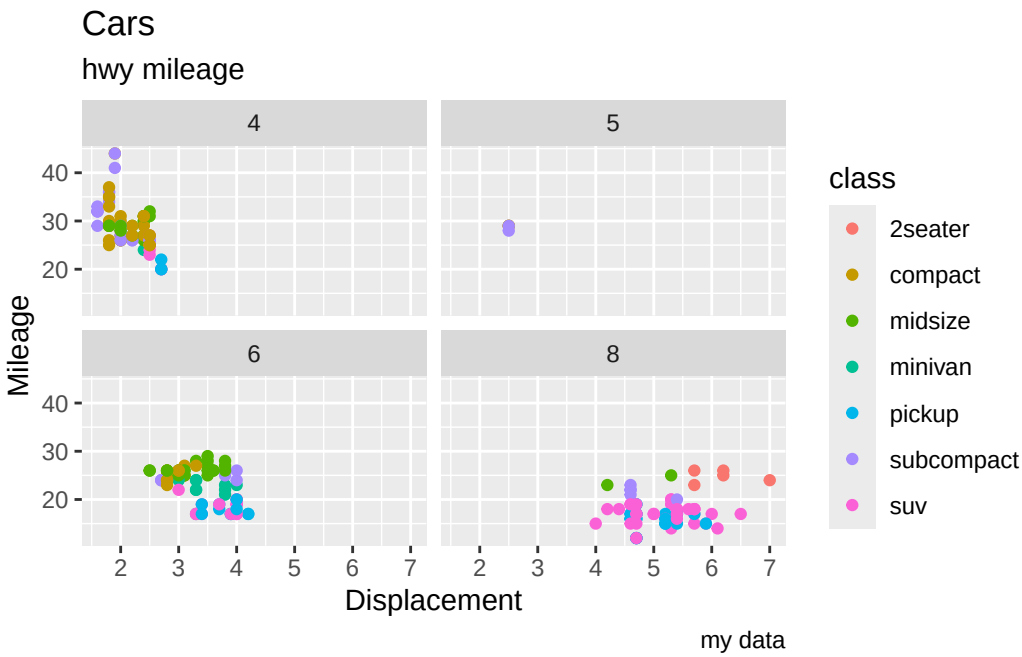
```
[1] 2 3 4 5 6
```

Let's make a graph

```
### ggplot2 is downloaded
library(ggplot2)
ggplot(cars) + aes(x=speed, y=dist)+ geom_point(col="BLUE")
```

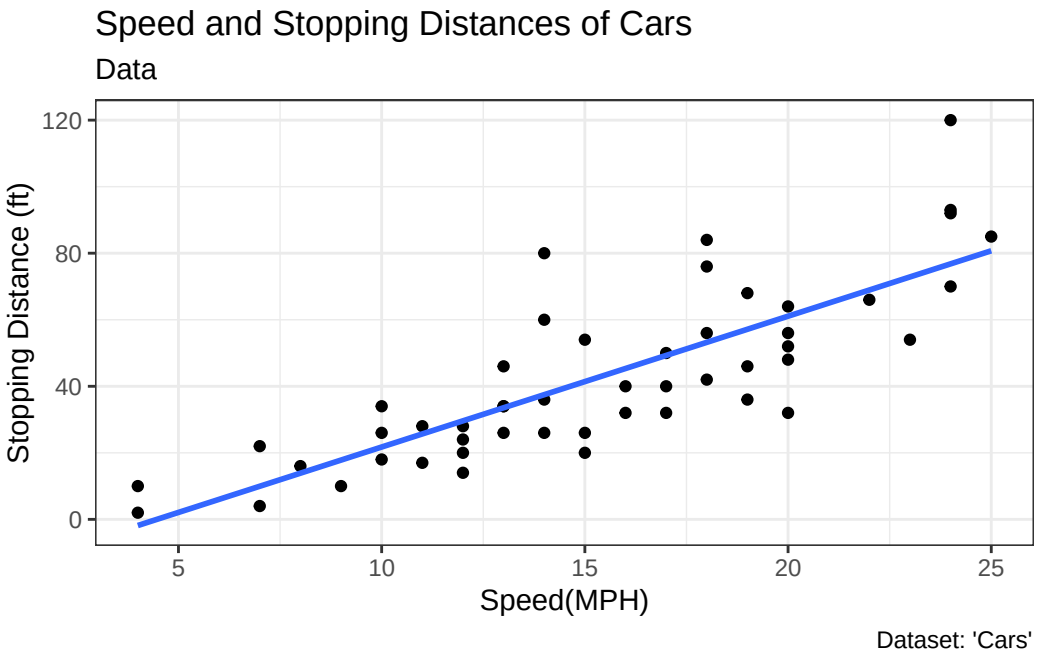



```
# Plots

library(ggplot2)

ggplot(mpg) +aes(x=displ, y=hwy, color=class) + geom_point()+ theme_gray() +labs(title="Cars
```




```r
# Lab 05 Assignment
library(ggplot2)

ggplot(cars) + aes(x=speed, y=dist) + geom_point()+ geom_smooth(method="lm", se=FALSE) + labs
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Q1: Which geometric layer should be used to create scatter plots in ggplot2?

Answer: geom_point()

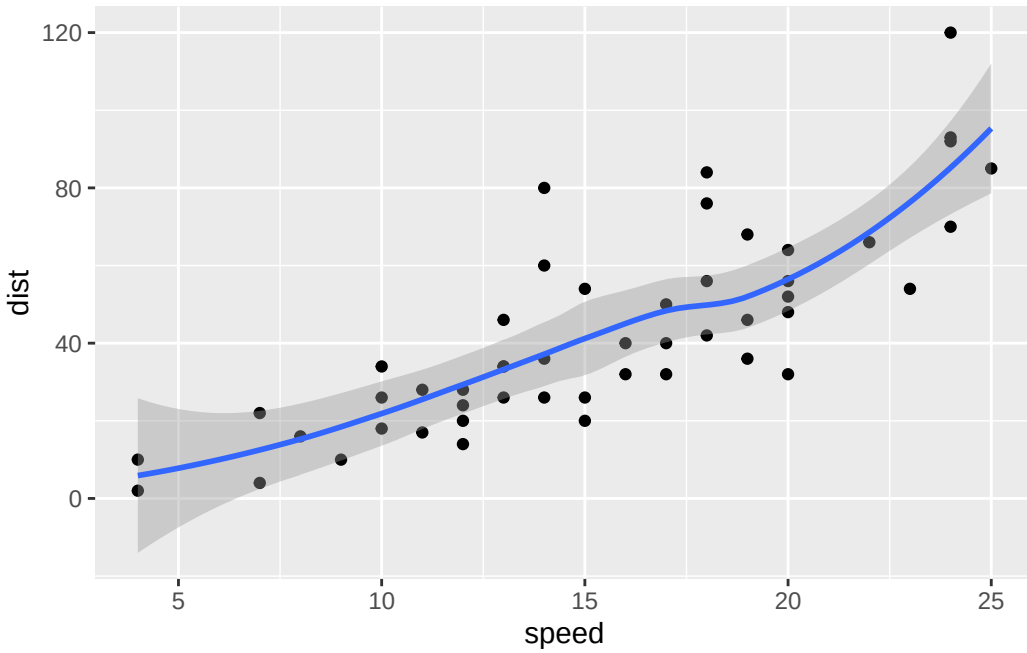
Q2:In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the geom_smooth() function?

Answer: Is added to the plot

Note: Please check plot/code below of cars dataset with geom_smooth()

```
library(ggplot2)
ggplot(cars) + aes(x=speed, y=dist) + geom_point()+ geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



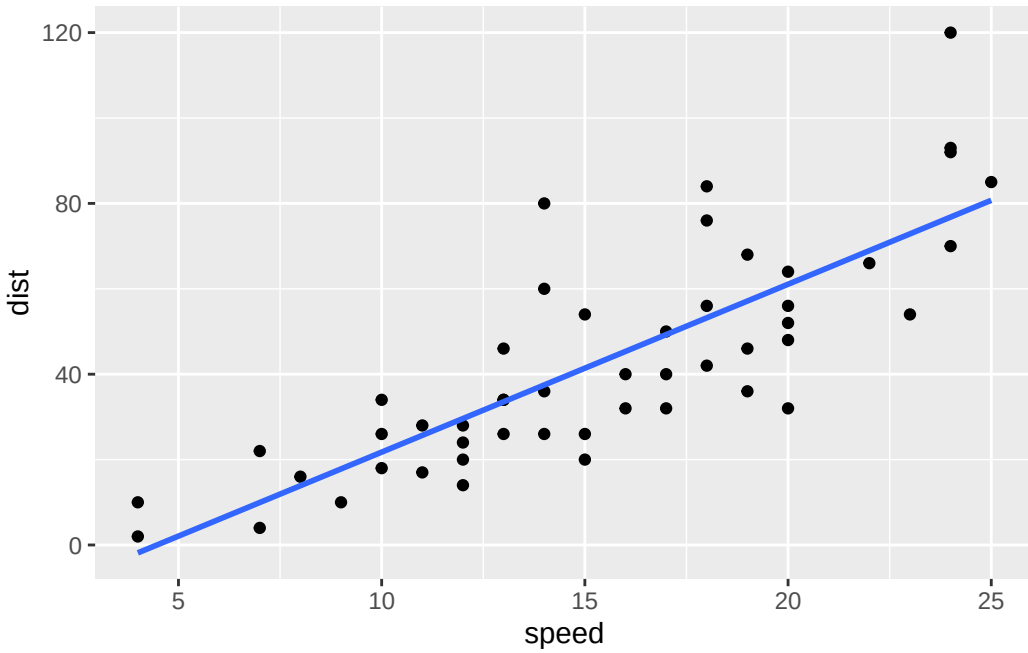
Q3: Argue with geom_smooth() to add a straight line from a linear model without the shaded standard error region?

Answer: Is added to the plot

Note: Please check plot/code below that shows plot of dataset with a straight line.

```
library(ggplot2)

ggplot(cars) + aes(x=speed, y=dist) + geom_point()+ geom_smooth(method="lm", se=FALSE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



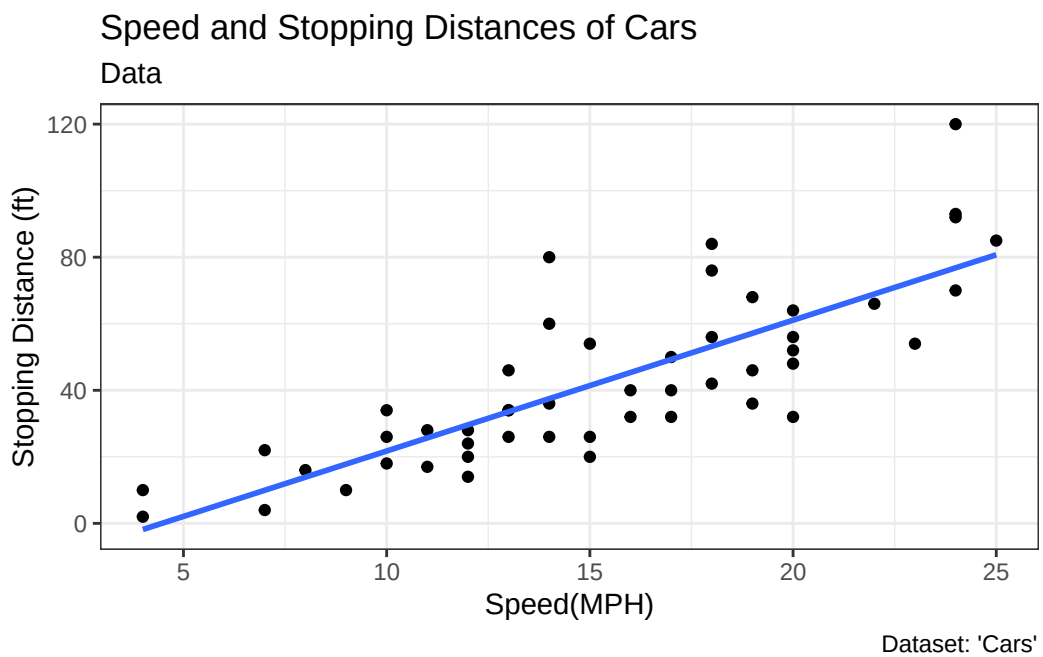
Q4: Can you finish this plot by adding various label annotations with the labs() function and changing the plot look to a more conservative "black & white" theme by adding the theme_bw() function:

Answer: Is added to the plot

Note: please check plot/code below of full plot with labels.

```
library(ggplot2)

ggplot(cars) + aes(x=speed, y=dist) + geom_point()+ geom_smooth(method="lm", se=FALSE) + labs
```

```
`geom_smooth()` using formula = 'y ~ x'
```



New section: Adding more plot aesthetics through aes

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Q5:Use the nrow() function to find out how many genes are in this dataset. What is your answer? Answer:5196

Check chunk below for code of # of rows for gene dataset

```r
nrow(genes)
```

```
[1] 5196
```

> **Q6: Use the colnames() function and the ncol() function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?**
>
> **Answer: There are 4 columns found and the column names are: "Genes", "Condition1", "Condition2" and "State"**

Check chunk below for code of # of columns

```r
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```r
ncol(genes)
```

```
[1] 4
```

> **Q7: Use the table() function on the State column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?**
>
> **Answer:127 upregulated genes**

Check chunk below for code of # of up regulated genes in data

```r
table(genes$State)
```

```
      down unchanging         up 
        72       4997        127 
```

> **Q8:Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset? Answer:2.44 of total genes are upregulated in dataset**

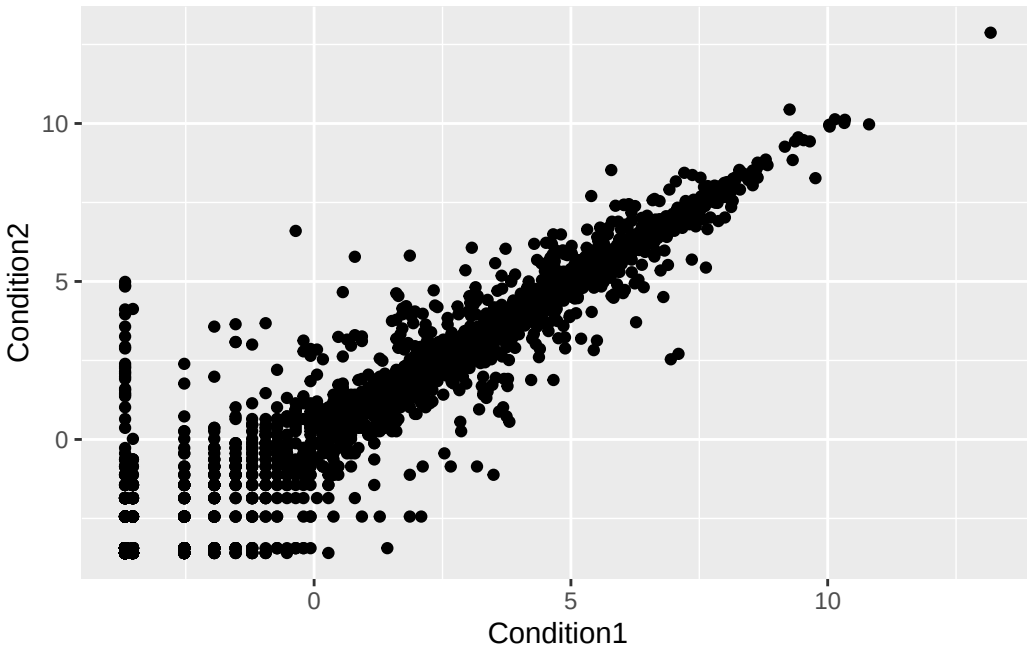
Check chunk below for code of total of up regulated genes in data

```r
round(table(genes$State)/nrow(genes)*100,2)
```

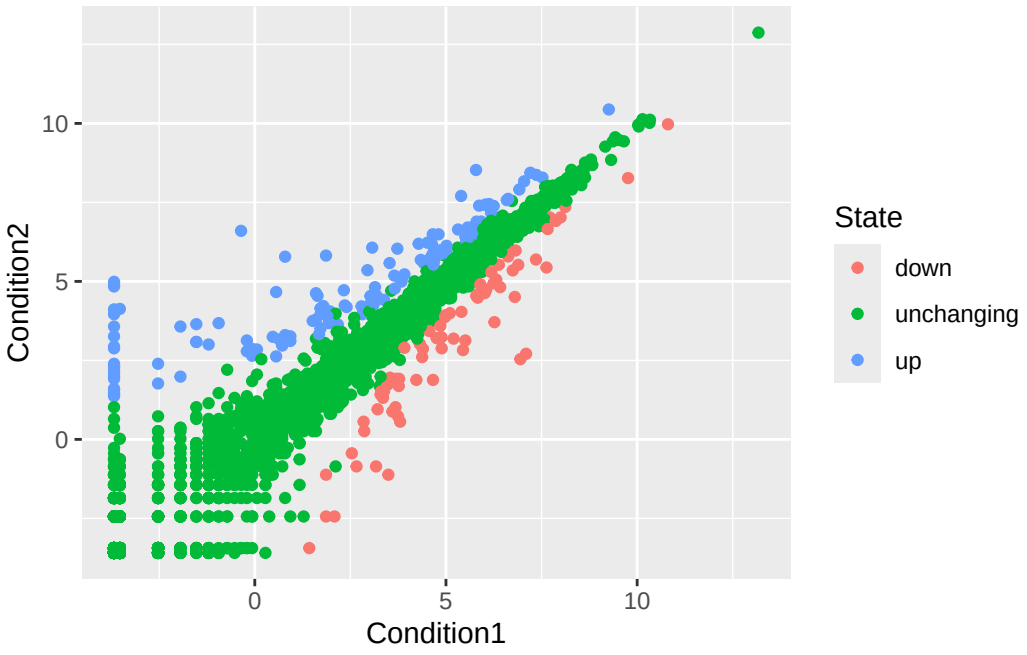
```
 down unchanging         up
 1.39      96.17       2.44
```

Q9. Complete the code below to produce the following plot ** Answer is in the r chunk below**

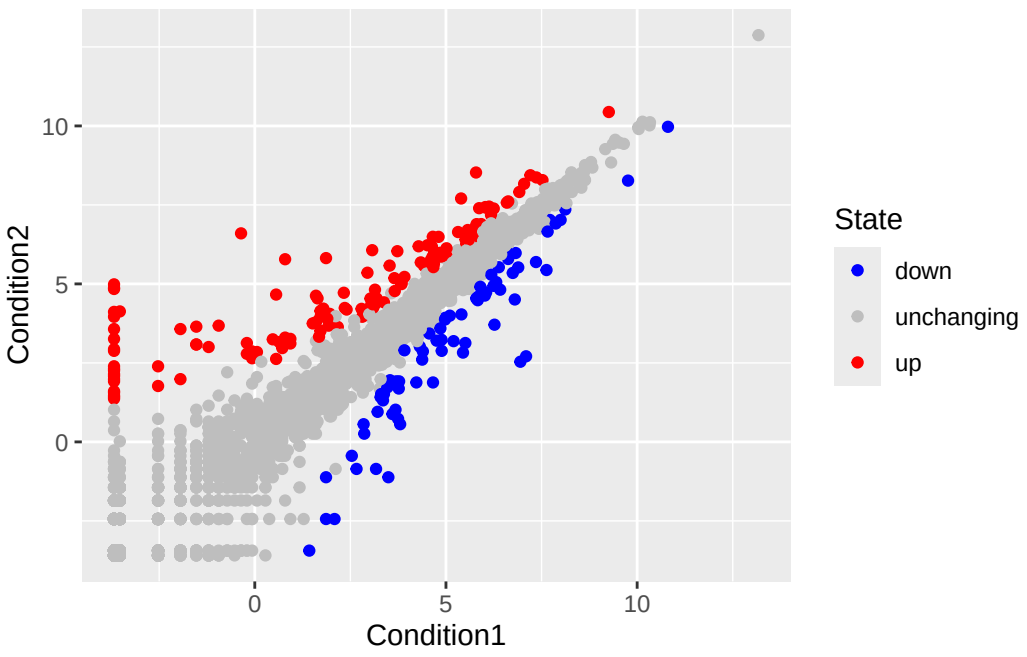
```r
ggplot(genes)+ aes(x=Condition1, y=Condition2) + geom_point()
```



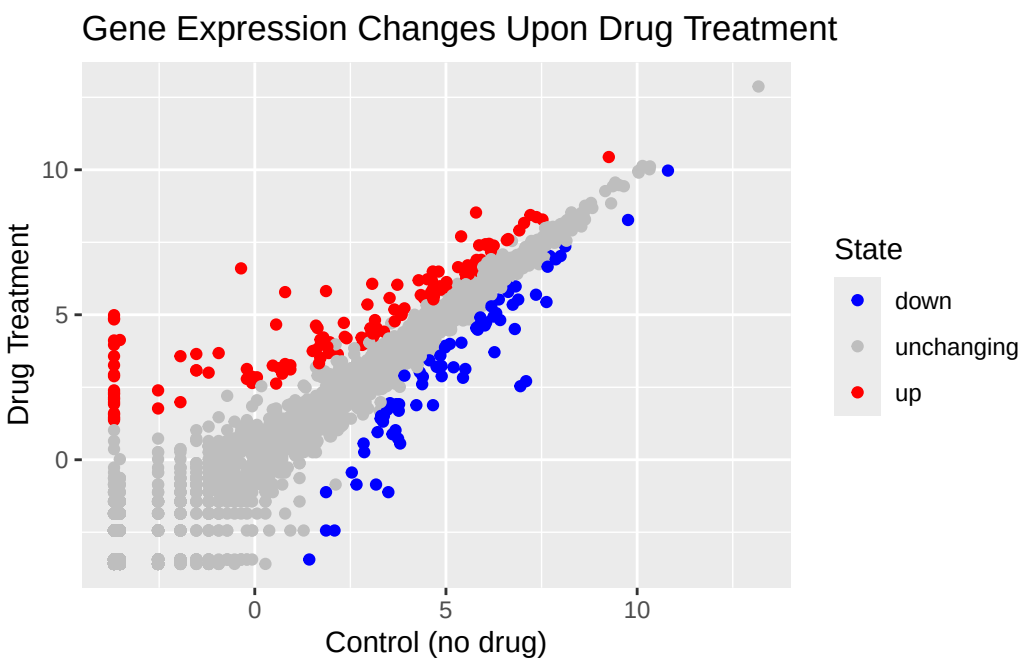

```r
p<- ggplot(genes) + aes(x=Condition1, y=Condition2, col=State) + geom_point()
p
```




```
p+ scale_color_manual( values=c("blue","gray", "red"))
```



** Q10. Nice, now add some plot annotations to the p object with the labs()

function so your plot looks like the following: ** Answer is shown below in the r chunk**

```
p+ scale_color_manual( values=c("blue","gray", "red"))+ labs(title="Gene Expression Changes U
```